

Problem Set #25

UMass Recreational Math Club

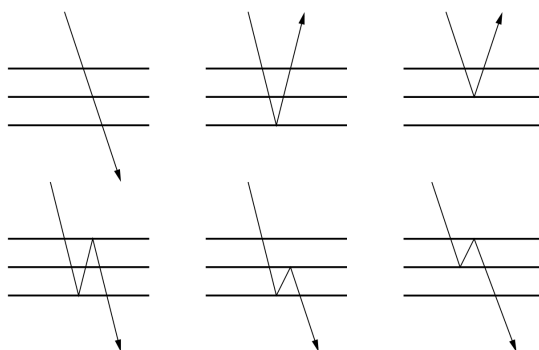
October 8th, 2025

Problem 1. What is the greatest integer that divides $p^4 - 1$ for all primes $p > 5$?

Problem 2. The numbers 1, 2, 3, 4, 5, 6, 7, 8, 9 are written in a 3×3 array of squares such that there is one number in each square, and if two numbers are consecutive then the squares they occupy share an edge. The numbers in the four corners add up to 18. What is the number in the center?

Problem 3. A beam of light is passing through two adjacent panes of glass. The beam can be reflected or it can pass through the glass. Let a_n be the number of paths the beam can take if there are n reflections. The figure shows that $a_0 = 1$, $a_1 = 2$, and $a_2 = 3$. (We do not count the reflection that bounces the beam off the top edge; the beam must enter the glass.)

- Determine a_3 and a_4
- Write a recurrence relation for a_n .



Problem 4. Let F_n denote the n th Fibonacci number. Show that for any prime p , the sequence of remainders of F_n when divided by p is periodic. When does the period divide $p - 1$?